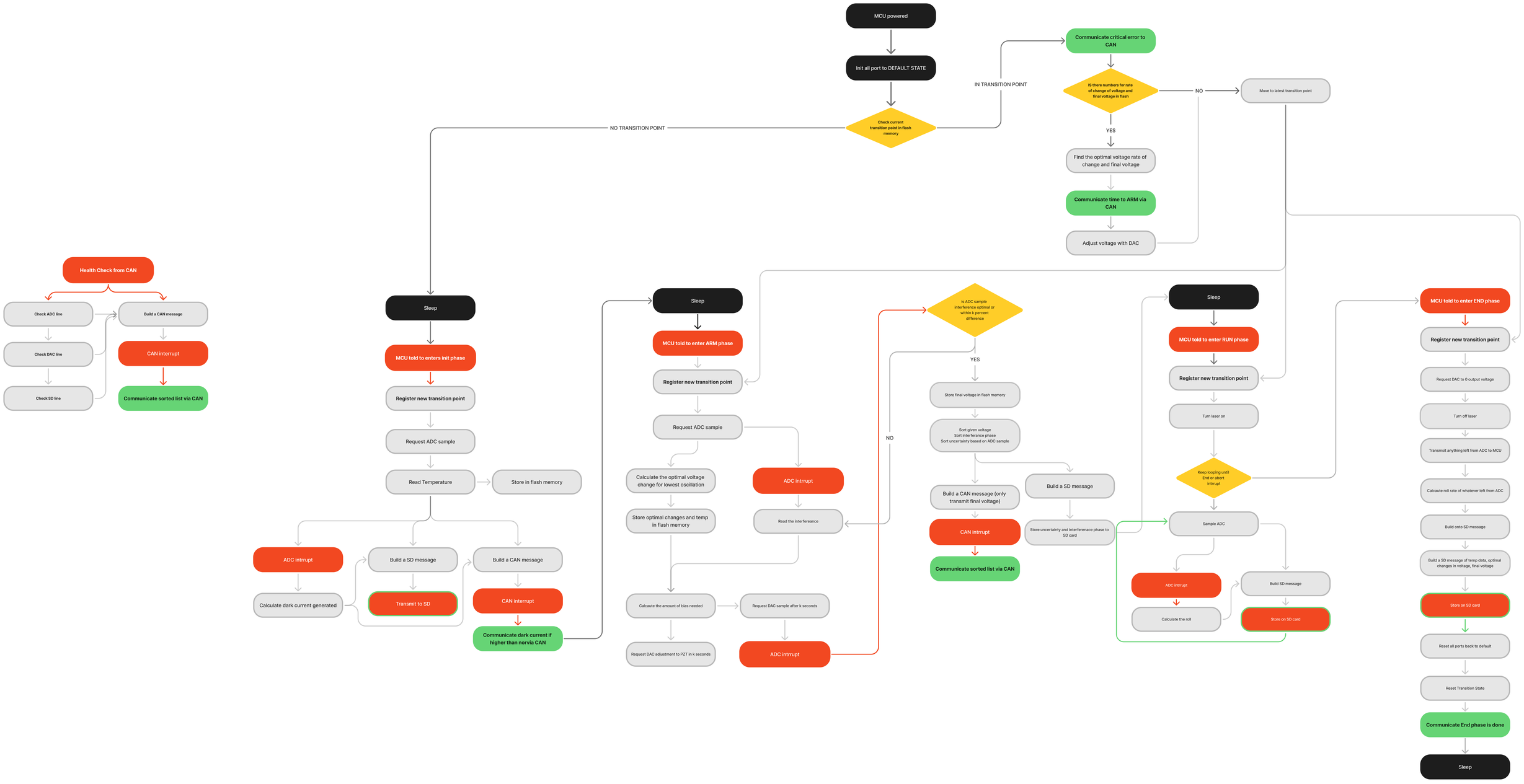
MCU powered
Init all port to DEFAULT STATE
Check current transition point in flash memory
IN TRANSITION POINT
NO TRANSITION POINT
Communicate critical error to CAN
IS there numbers for rate of change of voltage and final voltage in flash
NO
YES
Move to latest transition point
Find the optimal voltage rate of change and final voltage
Communicate time to ARM via CAN
Adjust voltage with DAC
Health Check from CAN
Check ADC line
Check DAC line
Check SD line
Build a CAN message
CAN interrupt
Communicate sorted list via CAN
Sleep
MCU told to enters init phase
Register new transition point
Request ADC sample
Read Temperature
Store in flash memory
ADC intrrupt
Calculate dark current generated
Build a SD message
Transmit to SD
Build a CAN message
CAN interrupt
Communicate dark current if higher than norvia CAN
Sleep
MCU told to enter ARM phase
Register new transition point
Request ADC sample
Calculate the optimal voltage change for lowest oscillation
Store optimal changes and temp in flash memory
Calcaute the amount of bias needed
Request DAC adjustment to PZT in k seconds
ADC intrrupt
Read the interfereance
Request DAC sample after k seconds
ADC intrrupt
Is ADC sample interference optimal or within k percent difference
NO
YES
Store final voltage in flash memory
Sort given voltage
Sort interferance phase
Sort uncertainty based on ADC sample
Build a CAN message (only transmit final voltage)
CAN intrrupt
Communicate sorted list via CAN
Build a SD message
Store uncertainty and interferenace phase to SD card
Sleep
MCU told to enter RUN phase
Register new transition point
Turn laser on
Keep looping until End or abort intrrupt
Sample ADC
ADC intrrupt
Calculate the roll
Build SD message
Store on SD card
MCU told to enter END phase
Register new transition point
Request DAC to 0 output voltage
Turn off laser
Transmsit anything left from ADC to MCU
Calcaute roll rate of whatever left from ADC
Build onto SD message
Build a SD message of temp data, optimal changes in voltage, final voltage
Store on SD card
Reset all ports back to default
Reset Transition State
Communicate End phase is done
Sleep